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5.1 EQUIPMENT SURVEY

INITIAL SETUP:

Tools and Special Tools

Site Calibration Kit 920-00636
Toolkit 730-00256
Flight test monitor kit

References

5 System Configuration WP
Theodolite COTS Manual

Materials/Parts**Equipment Condition**

TLS system assembled and inspected

Personnel Required

Technician (2)

WARNING

Coordinate with the airport authority to close the runway for the time required since work will be carried out on and around the runway.

NOTE

- The Maintenance Interface (MI) survey data entry screen requires the units to be in feet. The theodolite should be configured to display distance units in feet. It is not permitted to measure the distance in meters and then use a calculator in Windows or calculator phone app to convert to feet. To reduce the possibility of human error, the theodolite must be used in units of feet.
- Record verifications on a notepad with a pen or paper or in Table 1-1.
- Reference theodolite manufactures operator's manual for detailed instructions

GENERAL INFORMATION

After the TTLS sites have been installed, the ASA, ESA, ATA, Cal/BIT and Interrogation antenna must be surveyed. The survey is conducted relative to the runway threshold, the actual latitude and longitude for each point is not needed to operate TTLS, the latitude and longitude is only needed for requests to the regulator to place the equipment on airport property. The survey accuracy should be within 3 feet (1 meter) to honor the Lat/long indicated on the request form. The TTLS calibration process (Section 6.2 Calibration WP) eliminates the error of the survey.

TLS Equipment Survey

1. Configure Theodolite as follows:

NOTE

Ensure theodolite battery is charged prior to configuration.

- a. Place Theodolite at the centerline, threshold point of the runway.
- b. Level the Theodolite.
- c. Power on the theodolite.

- d. Verify unit of measure is in feet, adjust as needed.
2. Set up Electronic Distance Meter for distance measuring on Theodolite.
 - a. Adjust height of prism pole by placing pole on pavement a few inches away from Theodolite eyepiece and setting height to be within one inch of eyepiece height. Lock prism pole at this height and DO NOT change throughout the complete final precision survey procedure.
3. Orient the Theodolite as follows:
 - a. Align the Theodolite eyepiece on the centerline of the runway and set the centerline as the zero-azimuth measurement.
 - b. Align the Theodolite with the TTLS coordinate system by rotating the theodolite 180°, read from theodolite, and re-zero the azimuth (See Figure 1-1).

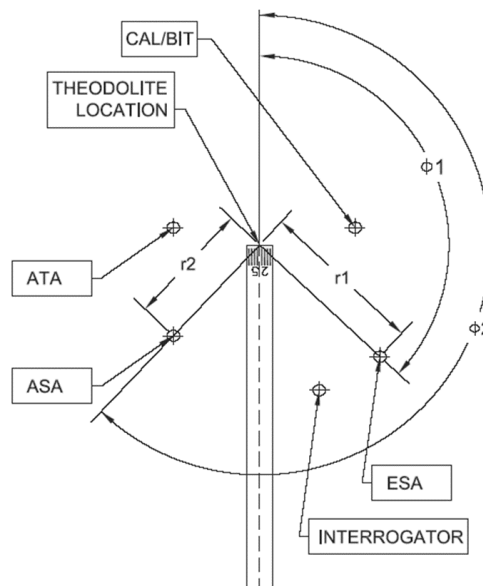


Figure 1-1. Theodolite Positioning

4. Survey TTLS sites as follows:
 - a. Technician 2 will hold the prism pole next to each antenna between antenna and theodolite.
- NOTE**
- Use the reference antenna for ASA and ESA.
 - Use two-way radio to facilitate communication between technicians.
- b. Technician 1 will use theodolite to measure azimuth, distance, and elevation to prism.
 - c. Record measurements in table 1-1 or note pad with pen.
5. Compute elevation measurement as follows:

- a. Record the raw measurement from the theodolite on a notepad with pen.
- b. If the reading from theodolite is $\approx 90^\circ$, Subtract the recorded measurement from 90°

(Example: $90^\circ - 89.1^\circ = 0.9^\circ$ or $90^\circ - 90.9^\circ = -0.9^\circ$)

- c. If the reading from theodolite is $\approx 270^\circ$, Subtract 270° from the recorded measurement

(Example: $270.9^\circ - 270^\circ = 0.9^\circ$ or $269.1^\circ - 270^\circ = -0.9^\circ$)

- d. This result is the elevation that must be entered in the MI survey data entry screen.

NOTE

- A positive result means the elevation is above the horizon and vice-versa for a negative result.
 - 1/10th increment enough accuracy for MI survey data entry.
6. Measure the distance of the high and low antenna from the reference antenna on the ASA and record in Table 1-1.

FTM Theodolite Position Survey

NOTE

- The ground elevation at threshold is likely to differ from the ground elevation at the FTM theodolite location. An approximate setback from threshold must be used to find the ground elevation difference and then the actual setback may be computed.
 - This procedure will establish the location for the Flight Test Monitor (FTM) theodolite that achieves a theodolite eyepiece height within a foot of the ideal glide path.
 - Provisions must be made in advance to have access to the runway threshold for 30 minutes.
1. Request permission from the airport manager or tower to approach the runway for 30 minutes of survey activity.
 2. Configure Theodolite for survey as follows:

NOTE

Ensure theodolite battery is charged prior to configuration.

- a. Place Theodolite at the centerline, threshold point of the runway.
- b. Level the Theodolite.

- c. Power on the theodolite.
 - d. Set up Electronic Distance Meter for distance measuring on Theodolite.
 - e. Verify unit of measure is in feet, adjust as needed.
3. Orient the Theodolite as follows:
 - a. Align the Theodolite eyepiece on the centerline of the runway and set the centerline as the zero-azimuth measurement.
 4. Use tables below to determine distance from threshold to hold prism pole.

2-3 Degrees

50 foot TCH		Glide Path Angle										
GS angle >		2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3
Eye Piece Height	54	1302.9	1240.9	1184.4	1132.9	1085.6	1042.1	1002.0	964.8	930.3	898.2	868.2
	56	1298.2	1236.3	1180.1	1128.7	1081.6	1038.3	998.3	961.3	926.9	894.9	865.0
	58	1293.4	1231.8	1175.7	1124.6	1077.6	1034.5	994.6	957.8	923.5	891.6	861.8
	60	1288.6	1227.2	1171.4	1120.4	1073.7	1030.7	991.0	954.2	920.1	888.3	858.7
	62	1283.9	1222.7	1167.0	1116.3	1069.7	1026.9	987.3	950.7	916.7	885.0	855.5
	64	1279.1	1218.1	1162.7	1112.1	1065.7	1023.0	983.6	947.2	913.3	881.7	852.3
	66	1274.3	1213.6	1158.4	1108.0	1061.7	1019.2	980.0	943.6	909.9	878.4	849.1
	68	1269.5	1209.0	1154.0	1103.8	1057.8	1015.4	976.3	940.1	906.5	875.2	845.9
	70	1264.8	1204.5	1149.7	1099.7	1053.8	1011.6	972.6	936.6	903.1	871.9	842.8
	72	1260.0	1199.9	1145.4	1095.5	1049.8	1007.8	969.0	933.0	899.6	868.6	839.6
	74	1255.2	1195.4	1141.0	1091.4	1045.8	1003.9	965.3	929.5	896.2	865.3	836.4
	76	1250.4	1190.9	1136.7	1087.2	1041.9	1000.1	961.6	925.9	892.8	862.0	833.2
	78	1245.7	1186.3	1132.3	1083.1	1037.9	996.3	957.9	922.4	889.4	858.7	830.0

2-3 Degrees

55 foot TCH		Glide Path Angle										
GS angle >		2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3
Eye Piece Height	54	1446.1	1377.2	1314.6	1257.3	1204.9	1156.6	1112.1	1070.8	1032.5	996.9	963.6
	56	1441.4	1372.7	1310.2	1253.2	1200.9	1152.8	1108.4	1067.3	1029.1	993.6	960.4
	58	1436.6	1368.1	1305.9	1249.0	1196.9	1149.0	1104.8	1063.8	1025.7	990.3	957.2
	60	1431.8	1363.6	1301.5	1244.9	1193.0	1145.2	1101.1	1060.2	1022.3	987.0	954.1
	62	1427.0	1359.0	1297.2	1240.7	1189.0	1141.4	1097.4	1056.7	1018.9	983.7	950.9
	64	1422.3	1354.5	1292.9	1236.6	1185.0	1137.6	1093.7	1053.2	1015.5	980.4	947.7
	66	1417.5	1349.9	1288.5	1232.4	1181.0	1133.7	1090.1	1049.6	1012.1	977.1	944.5
	68	1412.7	1345.4	1284.2	1228.3	1177.1	1129.9	1086.4	1046.1	1008.7	973.9	941.3
	70	1407.9	1340.8	1279.8	1224.1	1173.1	1126.1	1082.7	1042.6	1005.3	970.6	938.2
	72	1403.2	1336.3	1275.5	1220.0	1169.1	1122.3	1079.1	1039.0	1001.9	967.3	935.0
	74	1398.4	1331.8	1271.2	1215.8	1165.1	1118.5	1075.4	1035.5	998.5	964.0	931.8
	76	1393.6	1327.2	1266.8	1211.7	1161.2	1114.6	1071.7	1032.0	995.1	960.7	928.6
	78	1388.9	1322.7	1262.5	1207.5	1157.2	1110.8	1068.1	1028.4	991.7	957.4	925.4

2-3 Degrees

60 foot TCH		Glide Path Angle										
GS angle >		2	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3
Eye Piece Height	54	1589.3	1513.6	1444.7	1381.8	1324.2	1271.2	1222.2	1176.9	1134.8	1095.6	1059.0
	56	1584.5	1509.0	1440.4	1377.7	1320.2	1267.3	1218.5	1173.3	1131.4	1092.3	1055.8
	58	1579.8	1504.5	1436.0	1373.5	1316.2	1263.5	1214.9	1169.8	1128.0	1089.0	1052.6
	60	1575.0	1499.9	1431.7	1369.4	1312.3	1259.7	1211.2	1166.3	1124.6	1085.7	1049.5
	62	1570.2	1495.4	1427.4	1365.2	1308.3	1255.9	1207.5	1162.7	1121.1	1082.4	1046.3
	64	1565.4	1490.8	1423.0	1361.1	1304.3	1252.1	1203.9	1159.2	1117.7	1079.1	1043.1
	66	1560.7	1486.3	1418.7	1356.9	1300.3	1248.3	1200.2	1155.7	1114.3	1075.8	1039.9
	68	1555.9	1481.8	1414.3	1352.8	1296.4	1244.4	1196.5	1152.1	1110.9	1072.6	1036.7
	70	1551.1	1477.2	1410.0	1348.6	1292.4	1240.6	1192.8	1148.6	1107.5	1069.3	1033.6
	72	1546.4	1472.7	1405.7	1344.5	1288.4	1236.8	1189.2	1145.1	1104.1	1066.0	1030.4
74	1541.6	1468.1	1401.3	1340.3	1284.4	1233.0	1185.5	1141.5	1100.7	1062.7	1027.2	
76	1536.8	1463.6	1397.0	1336.2	1280.4	1229.2	1181.8	1138.0	1097.3	1059.4	1024.0	
78	1532.0	1459.0	1392.6	1332.0	1276.5	1225.4	1178.2	1134.5	1093.9	1056.1	1020.8	

3-4 Degrees

50 foot TCH		Glide Path Angle										
GS angle >		3	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4
Eye Piece Height	54	868.2	840.1	813.8	789.1	765.9	743.9	723.2	703.6	685.0	667.4	650.7
	56	865.0	837.1	810.8	786.2	763.0	741.2	720.6	701.0	682.5	665.0	648.3
	58	861.8	834.0	807.9	783.3	760.2	738.5	717.9	698.4	680.0	662.5	645.9
	60	858.7	830.9	804.9	780.4	757.4	735.7	715.3	695.9	677.5	660.1	643.5
	62	855.5	827.8	801.9	777.6	754.6	733.0	712.6	693.3	675.0	657.6	641.1
	64	852.3	824.7	798.9	774.7	751.8	730.3	710.0	690.7	672.5	655.2	638.8
	66	849.1	821.7	795.9	771.8	749.0	727.6	707.3	688.1	670.0	652.7	636.4
	68	845.9	818.6	793.0	768.9	746.2	724.8	704.7	685.6	667.5	650.3	634.0
	70	842.8	815.5	790.0	766.0	743.4	722.1	702.0	683.0	665.0	647.9	631.6
	72	839.6	812.4	787.0	763.1	740.6	719.4	699.4	680.4	662.5	645.4	629.2
	74	836.4	809.4	784.0	760.2	737.8	716.7	696.7	677.8	659.9	643.0	626.8
	76	833.2	806.3	781.0	757.3	735.0	713.9	694.1	675.3	657.4	640.5	624.5
	78	830.0	803.2	778.1	754.4	732.2	711.2	691.4	672.7	654.9	638.1	622.1

3-4 Degrees

55 foot TCH		Glide Path Angle										
GS angle >		3	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4
Eye Piece Height	54	963.6	932.5	903.3	875.8	850.0	825.7	802.7	780.9	760.3	740.8	722.2
	56	960.4	929.4	900.3	872.9	847.2	822.9	800.0	778.3	757.8	738.3	719.8
	58	957.2	926.3	897.3	870.0	844.4	820.2	797.4	775.8	755.3	735.9	717.4
	60	954.1	923.2	894.3	867.2	841.6	817.5	794.7	773.2	752.8	733.4	715.0
	62	950.9	920.1	891.3	864.3	838.8	814.8	792.1	770.6	750.3	731.0	712.6
	64	947.7	917.1	888.4	861.4	836.0	812.0	789.4	768.0	747.8	728.5	710.3
	66	944.5	914.0	885.4	858.5	833.2	809.3	786.8	765.5	745.3	726.1	707.9
	68	941.3	910.9	882.4	855.6	830.4	806.6	784.1	762.9	742.7	723.6	705.5
	70	938.2	907.8	879.4	852.7	827.6	803.9	781.5	760.3	740.2	721.2	703.1
	72	935.0	904.8	876.4	849.8	824.8	801.1	778.8	757.7	737.7	718.8	700.7
	74	931.8	901.7	873.4	846.9	822.0	798.4	776.2	755.1	735.2	716.3	698.3
	76	928.6	898.6	870.5	844.0	819.2	795.7	773.5	752.6	732.7	713.9	696.0
	78	925.4	895.5	867.5	841.1	816.3	793.0	770.9	750.0	730.2	711.4	693.6

3-4 Degrees

60 foot TCH		Glide Path Angle										
GS angle >		3	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4
Eye Piece Height	54	1059.0	1024.8	992.7	962.5	934.2	907.4	882.1	858.2	835.6	814.1	793.7
	56	1055.8	1021.7	989.7	959.7	931.4	904.7	879.5	855.7	833.1	811.7	791.3
	58	1052.6	1018.6	986.7	956.8	928.6	902.0	876.8	853.1	830.6	809.2	788.9
	60	1049.5	1015.5	983.7	953.9	925.8	899.2	874.2	850.5	828.1	806.8	786.5
	62	1046.3	1012.5	980.8	951.0	922.9	896.5	871.6	847.9	825.6	804.3	784.2
	64	1043.1	1009.4	977.8	948.1	920.1	893.8	868.9	845.4	823.0	801.9	781.8
	66	1039.9	1006.3	974.8	945.2	917.3	891.1	866.3	842.8	820.5	799.4	779.4
	68	1036.7	1003.2	971.8	942.3	914.5	888.3	863.6	840.2	818.0	797.0	777.0
	70	1033.6	1000.2	968.8	939.4	911.7	885.6	861.0	837.6	815.5	794.5	774.6
	72	1030.4	997.1	965.9	936.5	908.9	882.9	858.3	835.0	813.0	792.1	772.2
	74	1027.2	994.0	962.9	933.6	906.1	880.2	855.7	832.5	810.5	789.7	769.9
	76	1024.0	990.9	959.9	930.7	903.3	877.4	853.0	829.9	808.0	787.2	767.5
	78	1020.8	987.8	956.9	927.9	900.5	874.7	850.4	827.3	805.5	784.8	765.1

- Technician 2 will pace off an approximate distance on runway centerline equal to the calculated result from the table.
- Technician 1 will sight toward the prism pole location.

7. Technician 2 adjust the placement of the prism pole until it is at the distance calculated from the table.
8. Mark this position for the FTM Calibration Test Flight (CTF). This procedure has resulted in measuring a position on the runway that places the theodolite eyepiece on the glide slope as shown in the figure below.

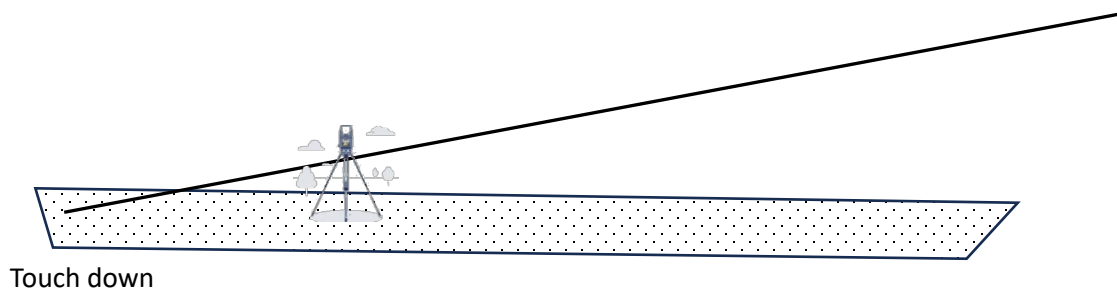


Figure 2 - Theodolite placed at 955 ft so that a 3.0 degree glide slope with TCH 55ft passes through the eyepiece

9. If an offset approach is required for calibration, determine position as follows:
 - a. With the theodolite still on the threshold centerline rotate theodolite 90° from runway centerline toward desired offset direction.
 - b. Technician 2 will hold prism pole and adjust placement at desired location until it is at 90°.
 - c. Technician 1 will record distance to prism pole measured from theodolite.
 - d. Mark this position with a stake or flag to use as a zeroing point when setting up theodolite for the CTF.
 - e. Move theodolite and prism pole to position determined in step 8.
 - f. Relevel and zero the theodolite.
 - g. Rotate theodolite 90° from runway centerline toward desired offset location.
 - h. Technician 2 will hold prism pole and adjust placement at desired location until it is at 90° and distance recorded in step 9.c.
 - i. Mark this position for the offset FTM CTF.

Mask Angle Measurement

A theodolite measurement of the ground obstructions or Mask Angle is required because the TTLS is a line-of-sight system (equipment needs to have an unobstructed "view" of the aircraft being tracked). A theodolite is used to measure +/- 45 degrees about the approach heading and measure the elevation of obstacles.

1. Set the theodolite up in front of the Interrogation antennas (within 5 feet).
2. Align the theodolite eyepiece on the stake used for alignment of the Interrogation antennas in Uplink Tower WP. Set this heading as the zero azimuth point on the theodolite.
3. Set the elevation of the theodolite and then record the lowest point at which sky is visible. Record this mask angle elevation for 0 degrees azimuth.
4. Rotate the theodolite slowly, measure and record the elevation mask angle every 5 degrees through ± 45 degrees from the runway heading. If there is an obstruction that peaks in between two 5° readings, measure and record the peak elevation and record to the nearest 5 degree azimuth point of the obstruction peak.
5. Record the defined mask angle and lowest unobscured azimuth space.
6. Calculate the lowest coverage altitudes at 10nm from apparent localizer location and record these altitudes in feet MSL.
7. This data will determine the Lowest Coverage Altitude during flight inspection.

END OF TASK

Post-Survey Checklist

1. ☐ Prism pole adjusted to theodolite eyepiece height
2. ☐ FTM Theodolite position surveyed and corresponds to the approach threshold crossing height
3. ☐ All structures are anchored properly
4. ☐ Archive previous data
5. ☐ Elevation recorded correctly (+/- sign)
6. ☐ Survey data entered into the system (rack A/B)
7. ☐ ASA low-res spacing entered as intended 20" or 30" to match actual antenna spacing
8. ☐ survey distances in units of feet, enter survey distances in units of feet
9. ☐ Survey data verified by a second person

- 10. ☐ Tune the F1 delay having jitter less than 4 ns for the ATA, ESA, and ASA
- 11. ☐ BIT stable with 10 cycles
- 12. ☐ Monitor limits set for ATA, ASA, ESA, Uplink 1-4, and Int
- 13. ☐ TOA calibrated using SysCal tool, entered on MI

END OF TASK

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